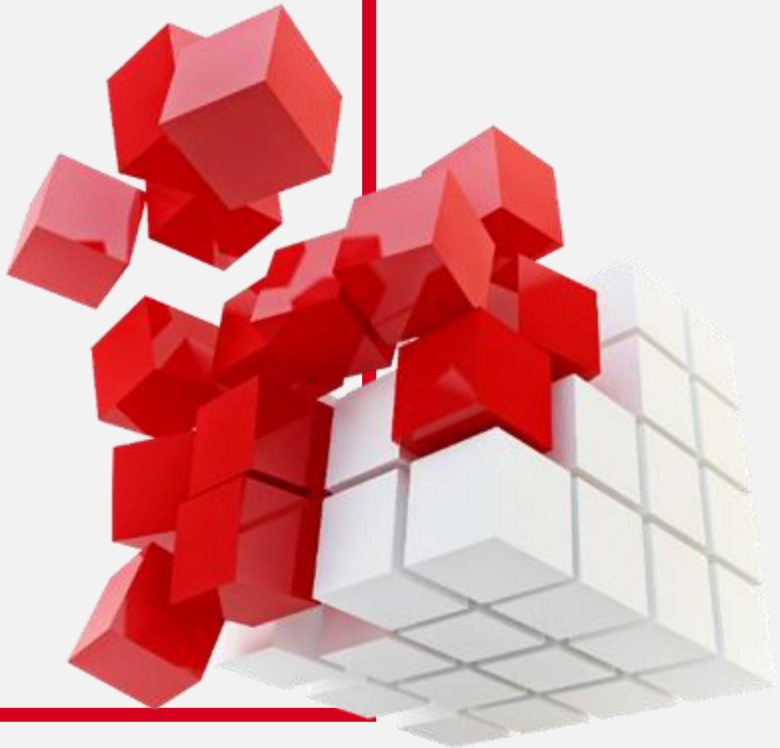
Build Agentic AI apps using Azure AI Agent Service



Phi Huynh

10th May 2025

Agenda

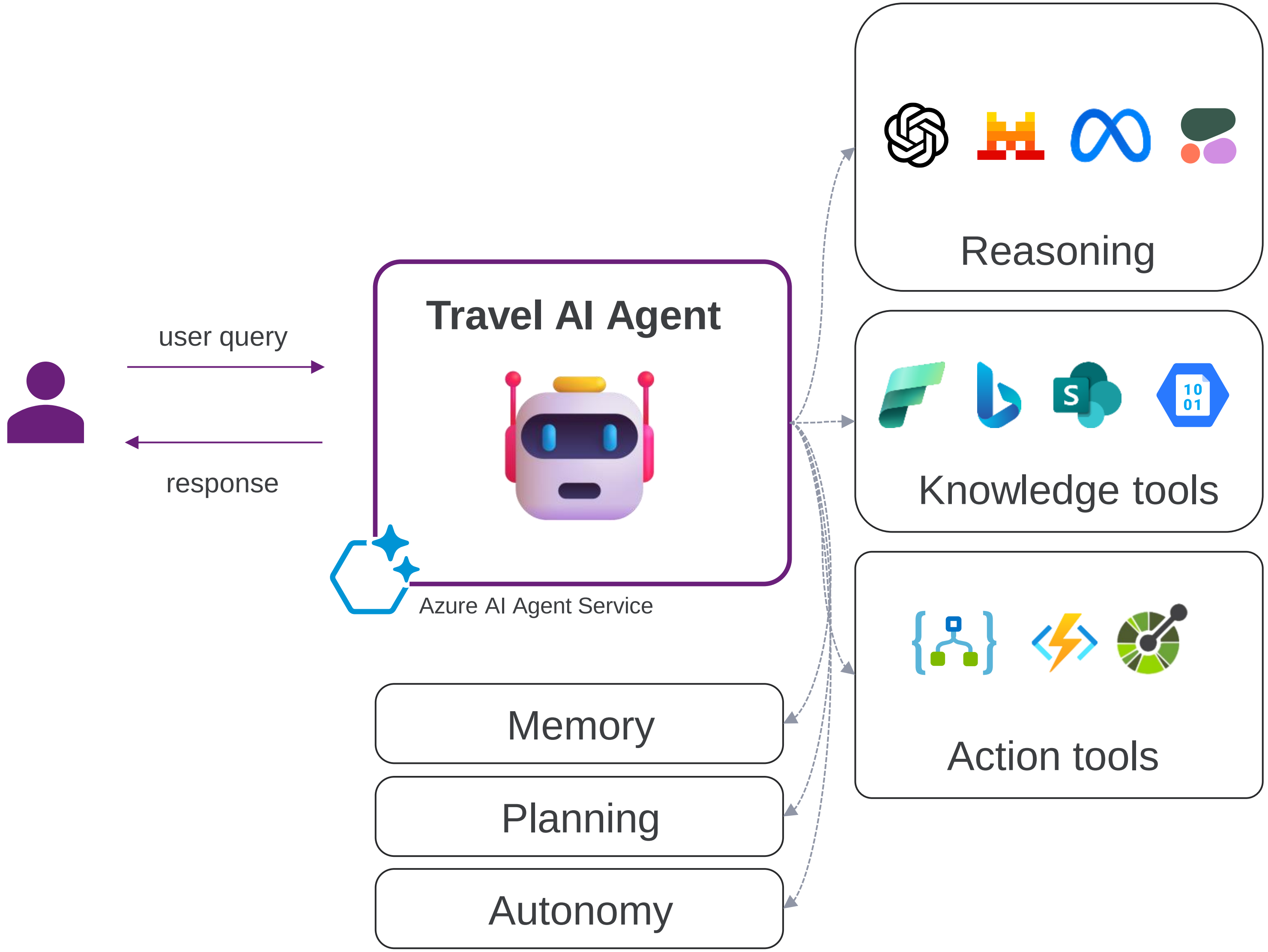
1. Azure AI Agents
2. Models (LLMs)
3. Knowledge tools
4. Action tools
5. Demo
6. Multi-agents

What is an AI Agent?

How an AI Agent works:

- Smart microservice with AI-powered
- Manage data (knowledge)
- Memory
- Out of the box tools
- Beyond chatbot
- Planning
- Autonomy
- Self-reflection & environment interaction
- Communicate with other agents
- …

See more about [Agentic AI]()



Options for agent development

DEV<Cafe
CONNECT AND SHARE

Autogen
Semantic
Kernel

OpenAI
Assistant
API

**Azure AI
Agent
Service**

Copilot
Studio

AG

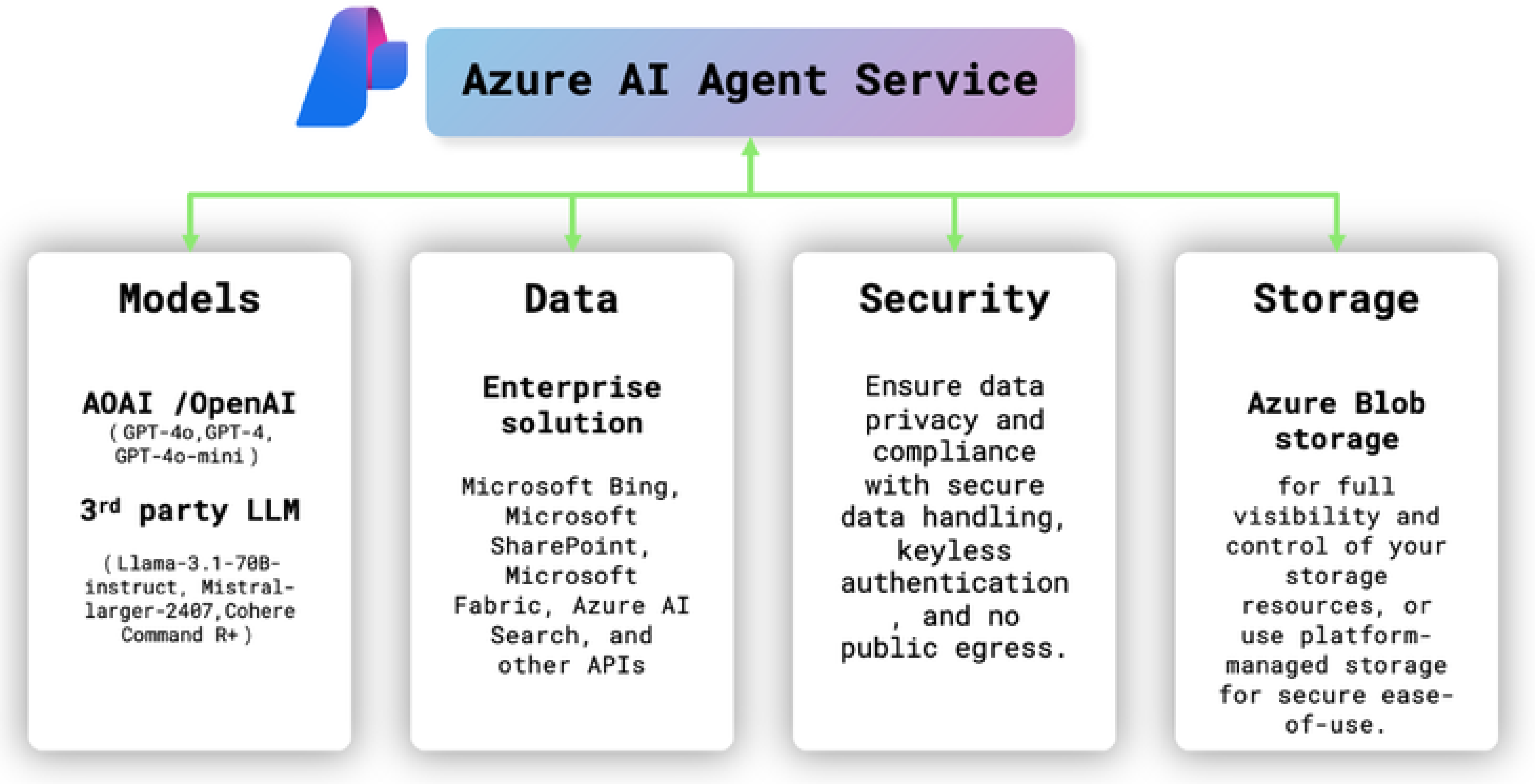
Azure AI Agent service

Deploy and scale AI agents with ease

Enterprise-grade Security	Extensive Data Connections	Flexible Model Selection	Rapid Development and Automation

ai.azure.com

Introducing Azure AI Agent service

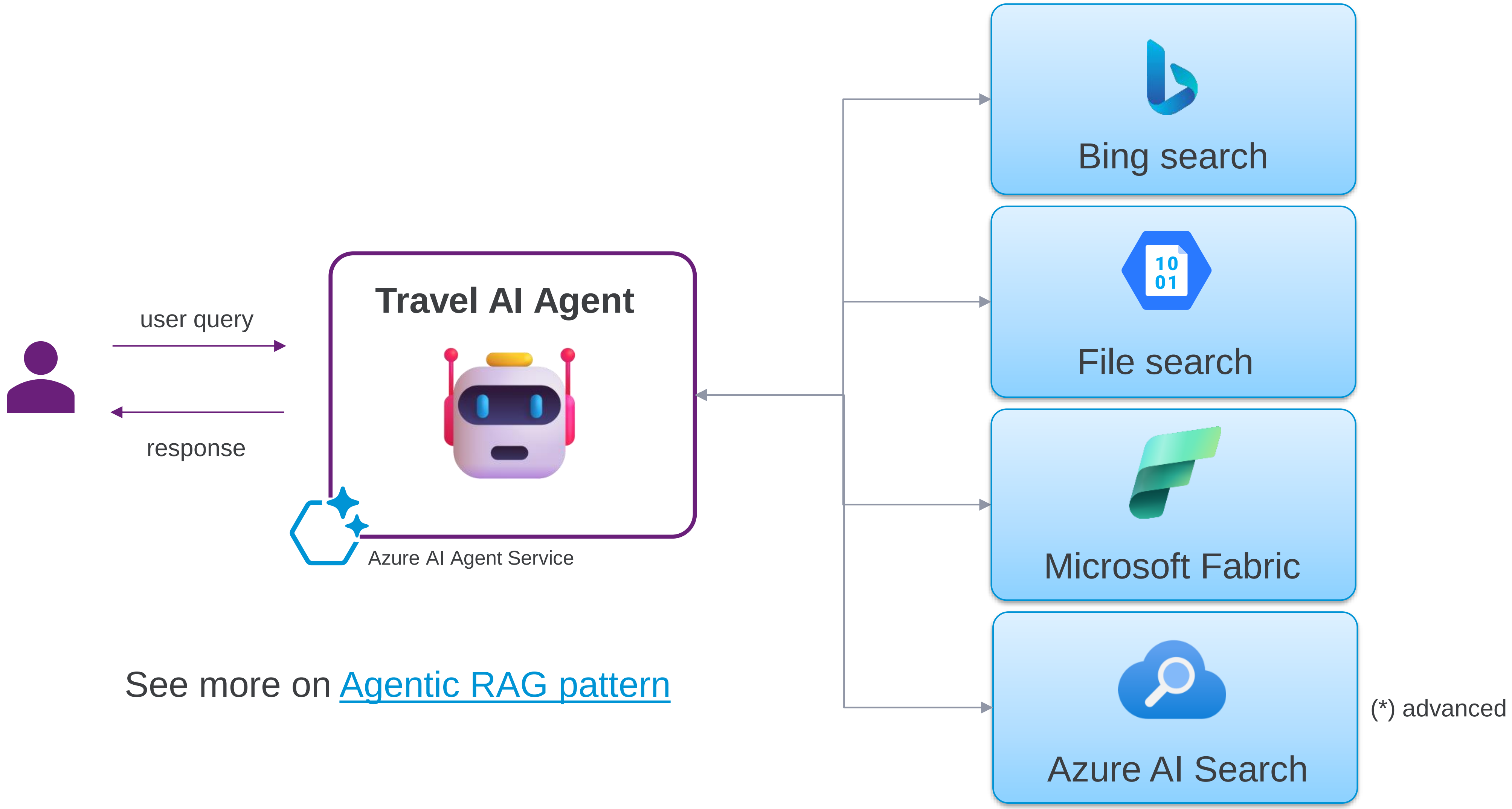


Supported models (both LLM & SLM)

Find the right model to build your custom AI solution

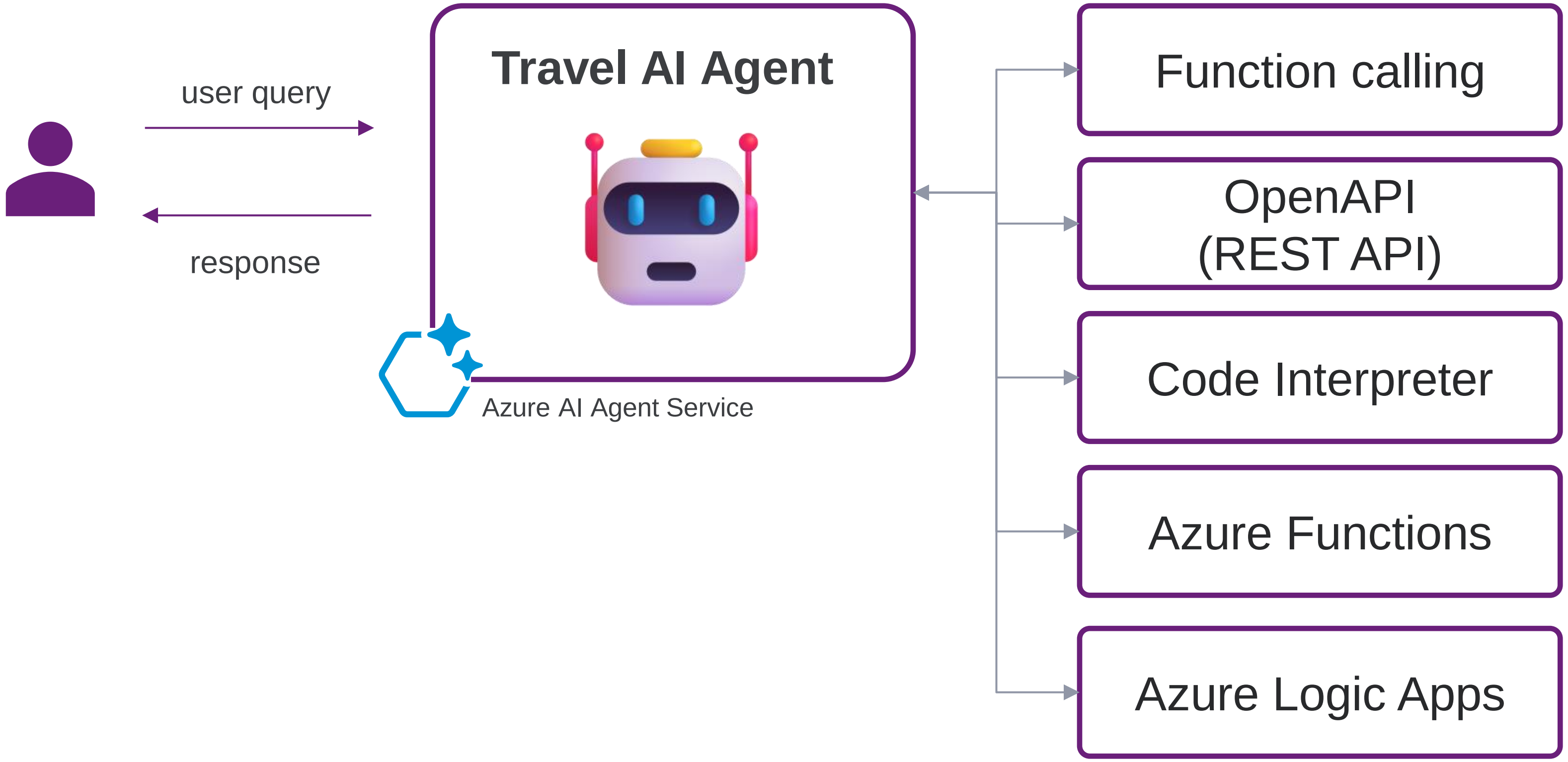
- **Phi-4-multimodal-instruct** — Chat completion
- **Phi-4** — Chat completion
- **mistral-small-2503** — Chat completion, Completions, C...
- **gpt-4o-mini-audio-preview** — Audio generation
- **gpt-4o-mini-realtime-preview** — Audio generation
- **o1** — Chat completion
- **o1-mini** — Chat completion
- **gpt-4o** — Chat completion
- **gpt-4o-mini** — Chat completion
- **gpt-4o-audio-preview** — Audio generation
- **gpt-4o-realtime-preview** — Audio generation
- **financial-reports-analysis-v2** — Chat completion
- **supply-chain-trade-regulati...** — Chat completion
- **Muse** — Image to image
- **Cohere-rerank-v3.5** — Text classification
- **Stable-Diffusion-3.5-Large** — Text to image, Image to image
- **Stable-Image-Ultra** — Text to image
- **Stable-Image-Core** — Text to image
- **Gretel-Navigator-Tabular** — Chat completion, Data generation
- **o1-preview** — Chat completion
- **Llama-3.3-70B-Instruct** — Chat completion
- **tsuzumi-7b** — Chat completion
- **Bria-2.3-Fast** — Text to image
- **Ministral-3B** — Chat completion
- **Virchow2** — image-feature-extraction
- **Prism** — Zero-shot image classification

Data (Knowledge) tools

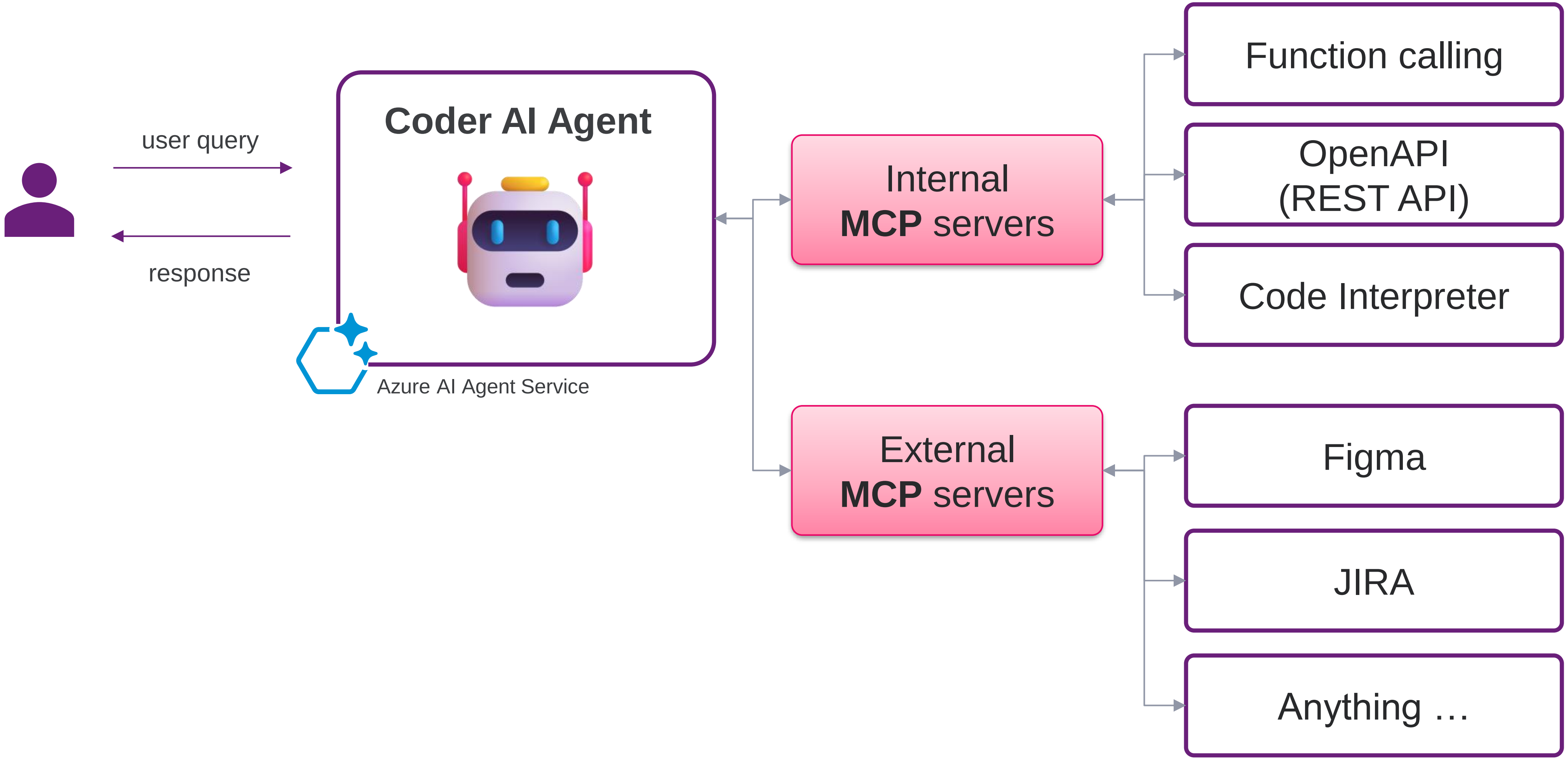


See more on [Agentic RAG pattern]

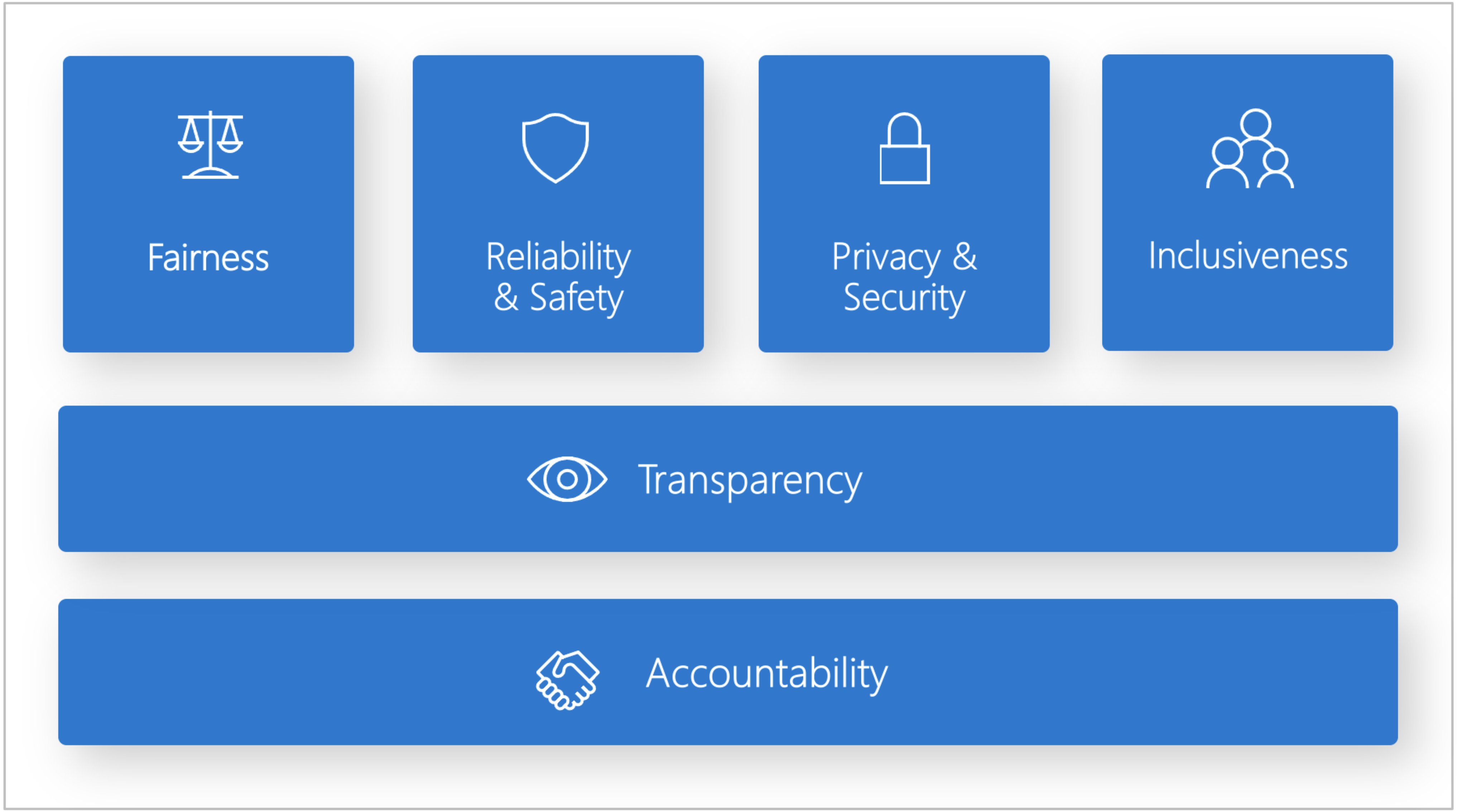
Action tools



Action tools & MCP



Responsible AI



Enterprise-grade Security

Tracing and monitoring

Content filters

Bring your own storage

Bring your own search index

On-behalf-of auth support

The full package

Azure AI Foundry – Agent Service

Built-in enterprise readiness

BYO-file storage

BYO-search index

OBO Authorization Support

Enhanced Observability

Extensive Ecosystem of Tools

Knowledge

- Microsoft Fabric
- SharePoint
- Bing Search
- Azure AI Search
- Your own licensed data
- Files (local or Azure Blob)
- File Search
- Code Interpreter

Actions

- Azure Logic Apps
- OpenAPI 3.0 Specified Tools
- Azure Functions

Model Catalog

Azure OpenAI Service

(GPT-4o, GPT-4o mini)

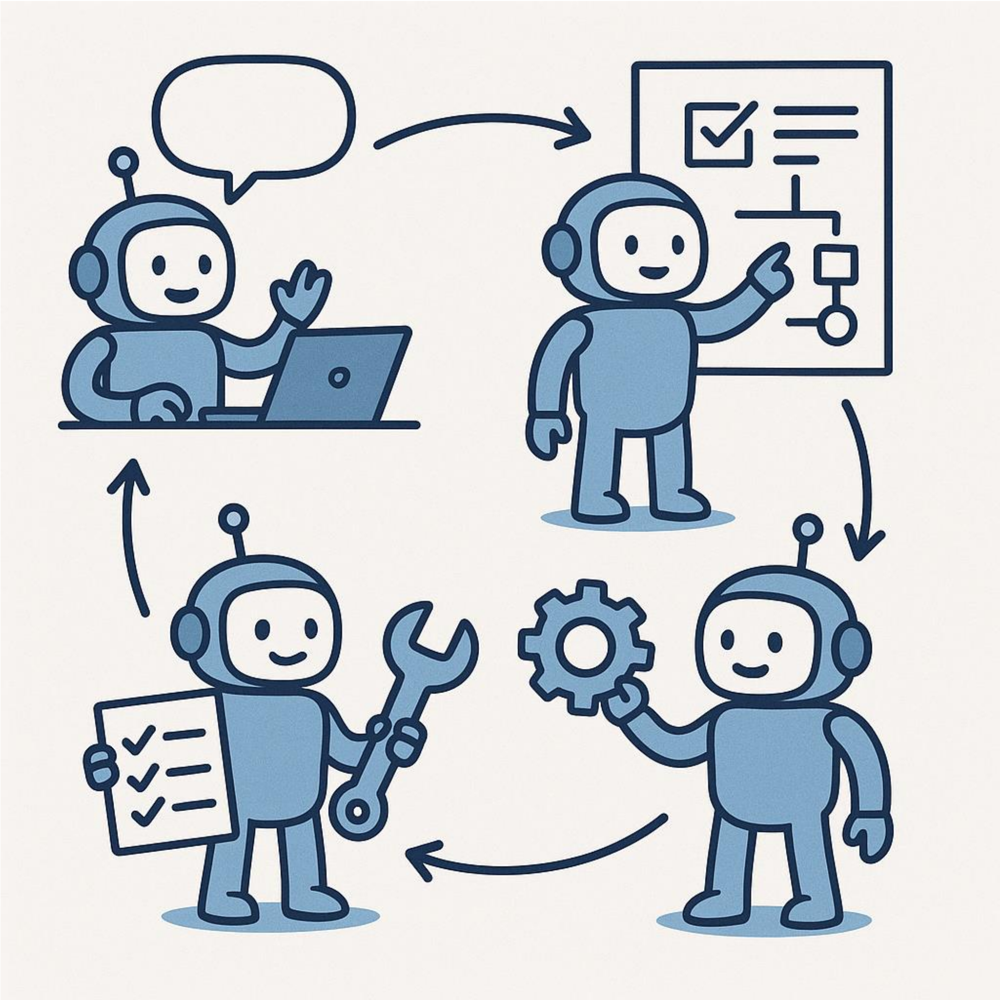
Models-as-a-Service

- Llama 3.1-405B-Instruct
- Mistral Large
- Cohere-Command-R-Plus

Demo – Implement an AI Agent on Azure

What you need

1. Azure AI Foundry
2. Deploy a model
3. Create Agent definition
   - Name & instructions
   - Model assignment
   - Tool/Skills
4. Run & test the agent
5. Integrate the Agent with your application



Real-world use cases for AI Agents

Personal productivity agent

Sales and Marketing support

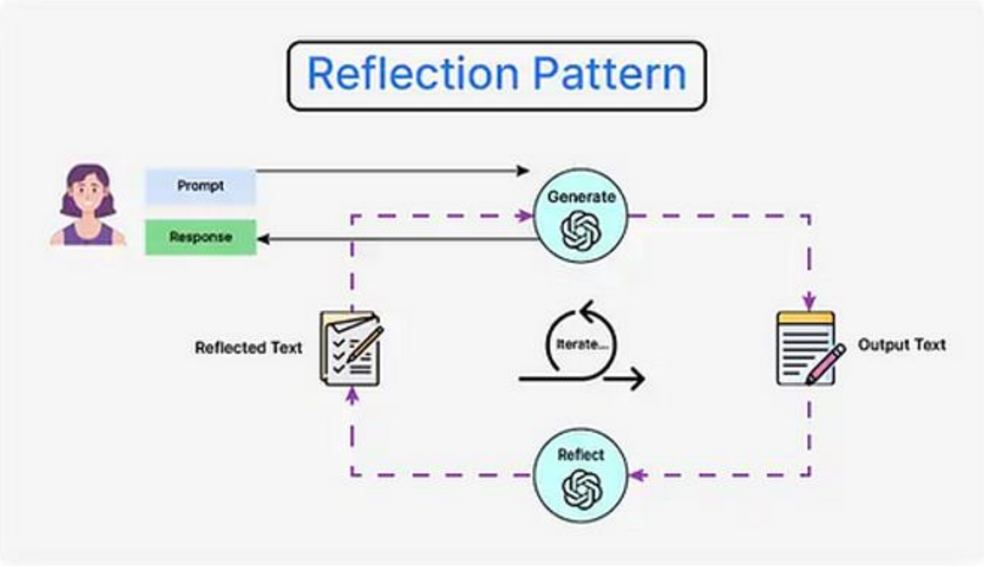
Customer service & IT helpdesk

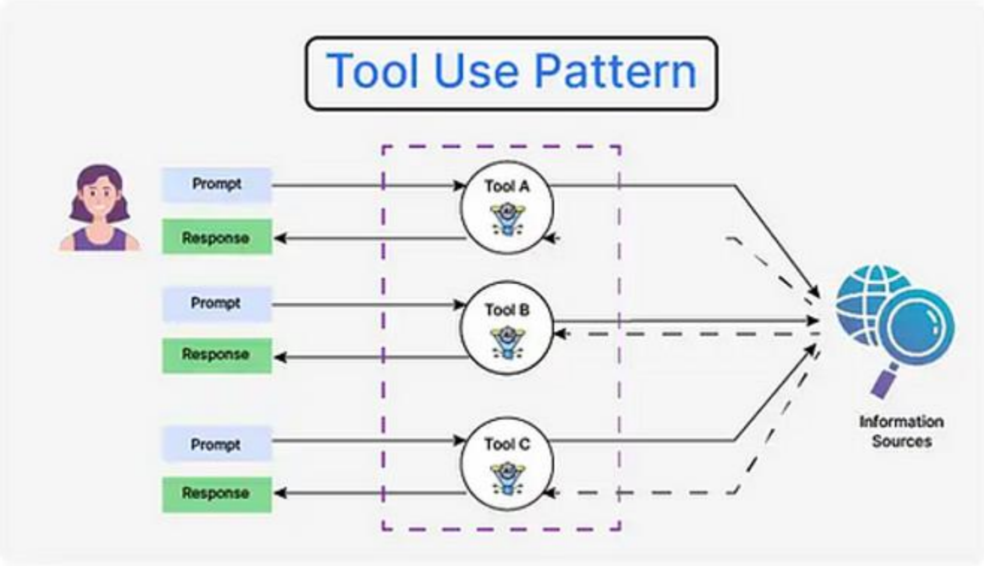
Developer assistant

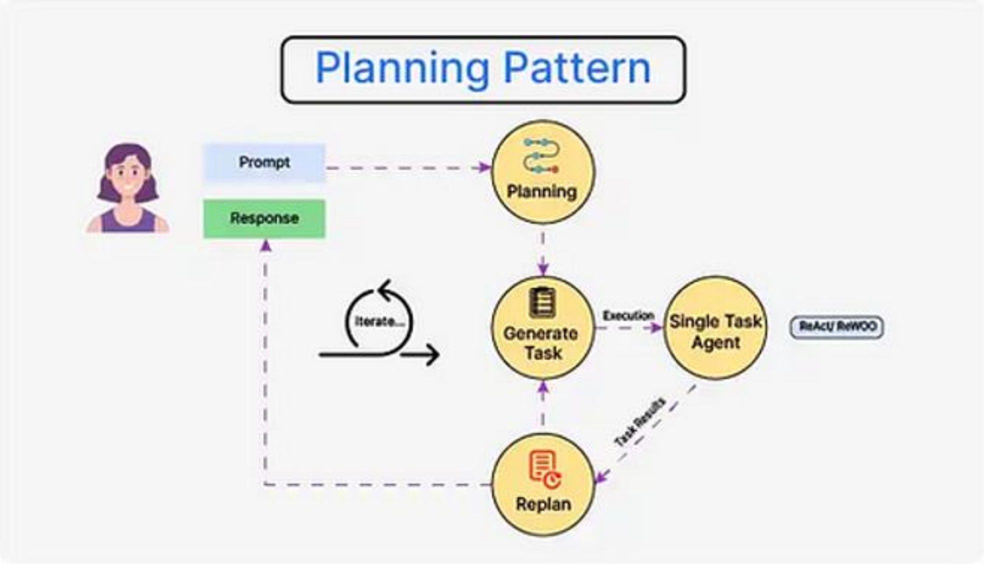
Search and knowledge management

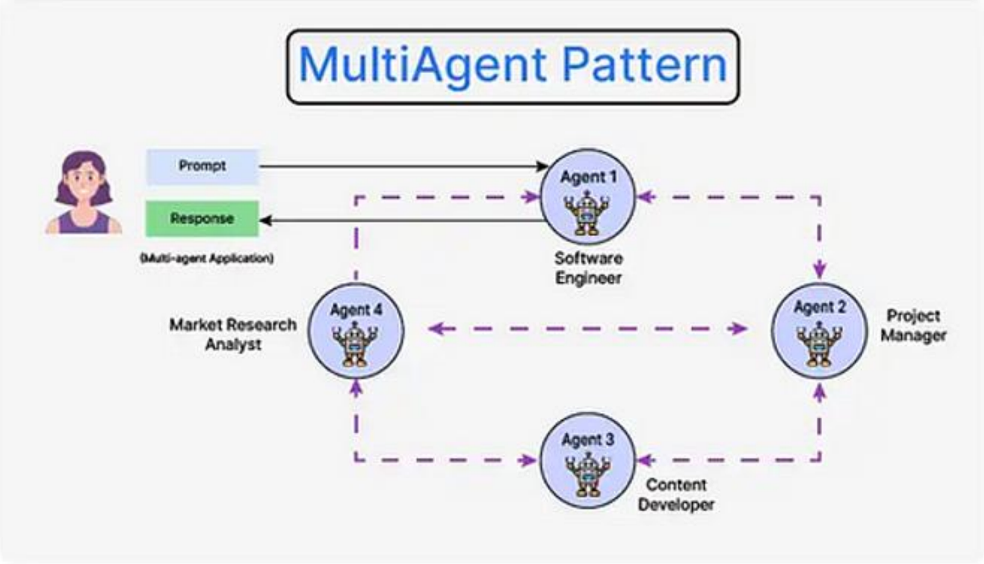
Operation optimization`

Agentic design patterns



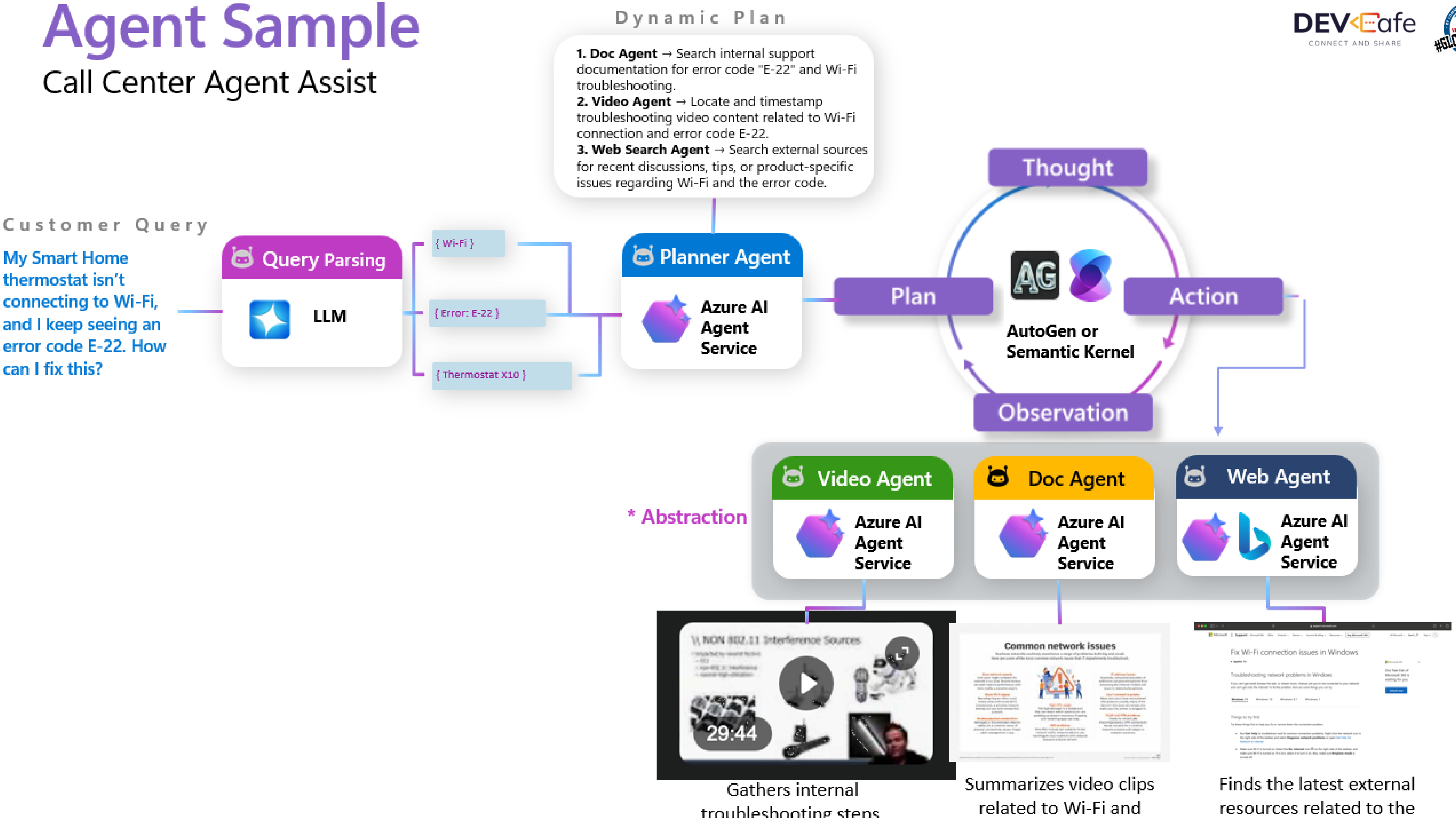




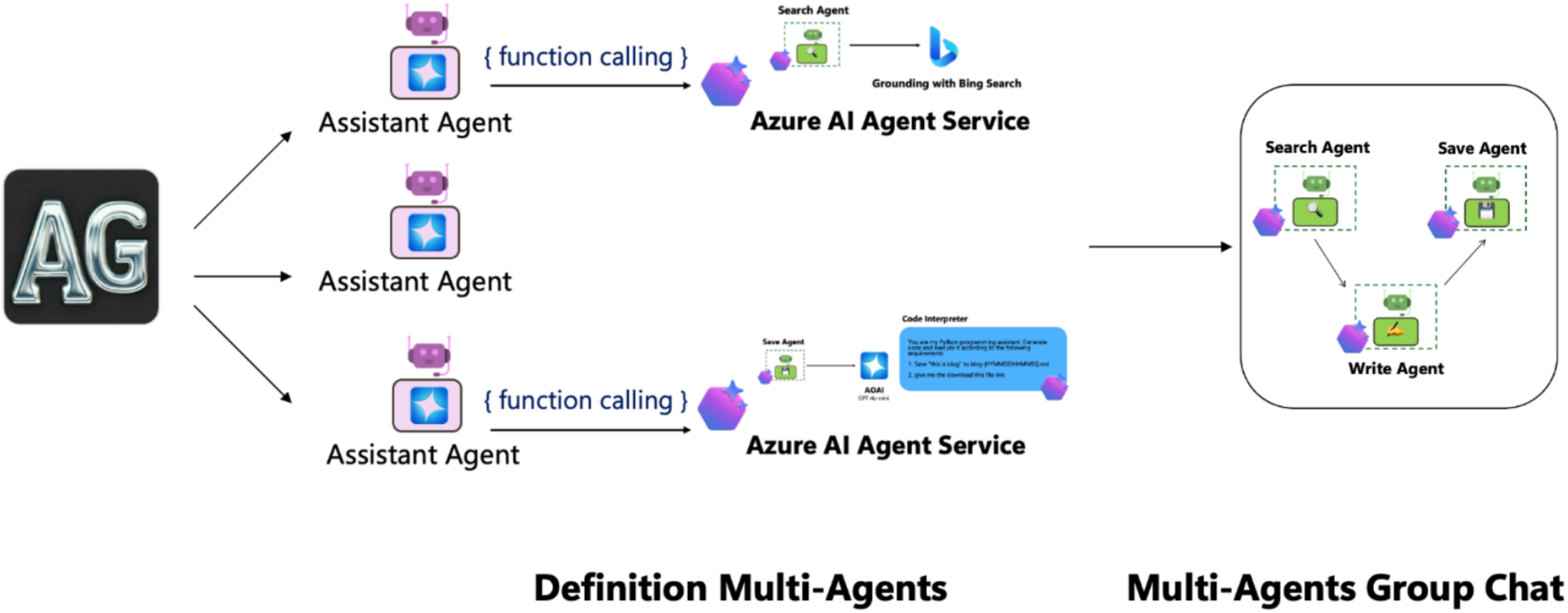


Agent Sample

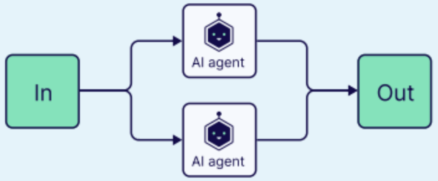
Call Center Agent Assist



Multi-agent orchestration

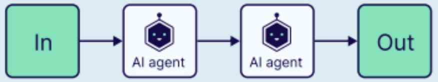


Multi-agent patterns



Parallel

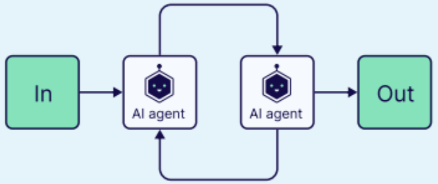
Multiple agents work simultaneously on different parts of a task.



Sequential

Tasks are processed sequentially, where one agent's output becomes the input for the next.

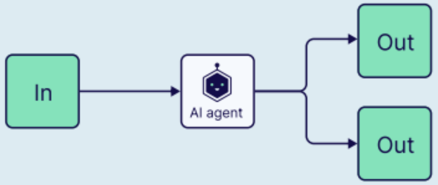
Example: Multi-step approvals.



Loop

Agents operate in iterative cycles, continuously improving their outputs based on feedback from other agent(s).

Example: Evaluation use cases, such as code writing and code testing.



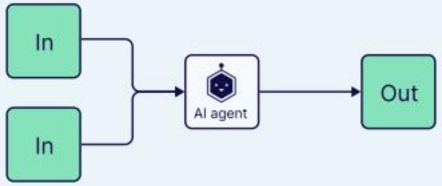
Router

A central router determines which agent(s) to invoke based on the task or input.

Aggregator

or synthesizer

Agents contribute outputs that are collected and synthesized by an aggregator agent into a final result.



Network

or horizontal

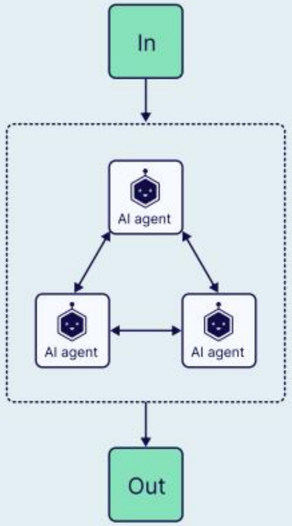
Agents communicate directly with one another in a many-to-many fashion, forming a decentralized network.

Pros:

- Distributed collaboration and group-driven decision-making.
- The system remains functional even if some agents fail.

Cons:

- Managing communication among agents can become challenging.
- More communication may cause inefficiencies and the possibility of agents duplicating efforts.



Hierarchical

or vertical

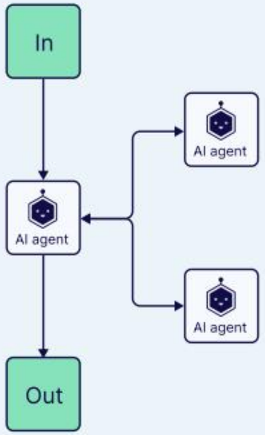
Agents are organized in a tree-like structure, with higher-level agents (supervisor agents) managing lower-level ones.

Pros:

- Clear division of roles and responsibilities among agents at different levels.
- Streamlined communication
- Suitable for large systems with a structured decision flow.

Cons:

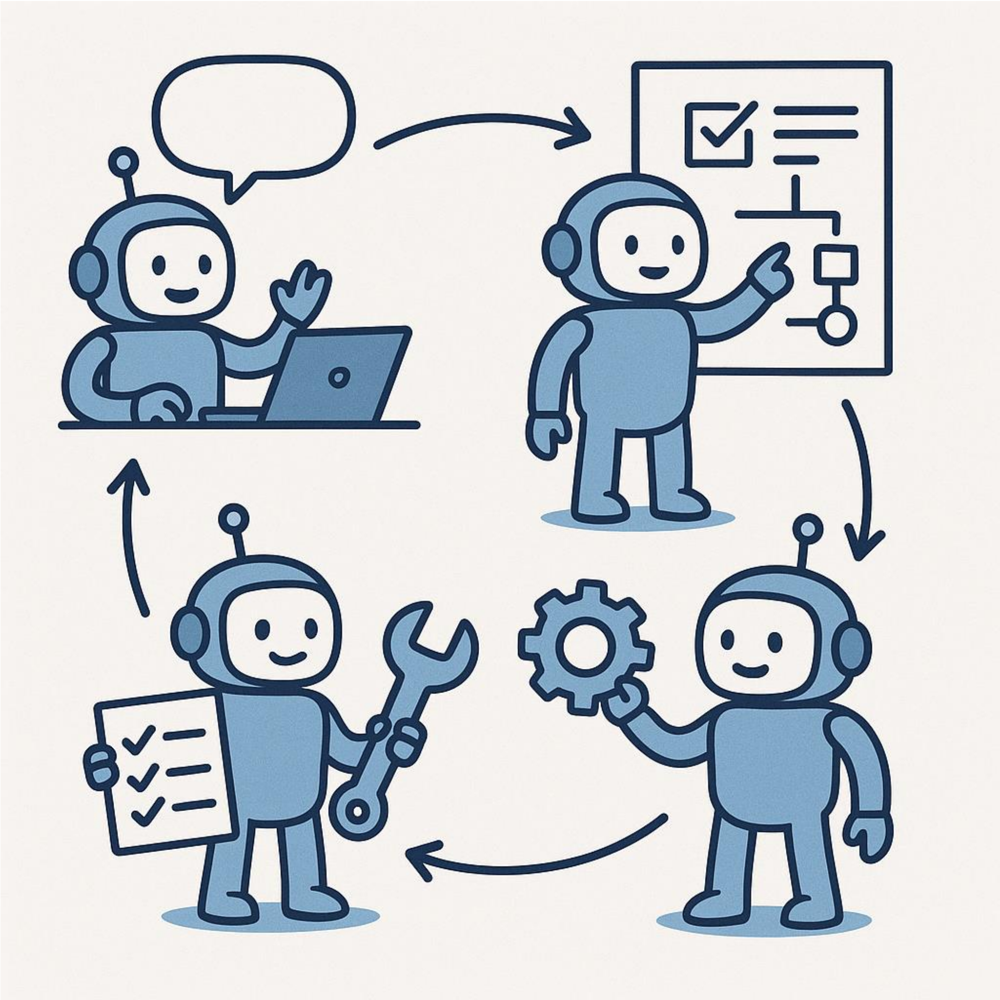
- Failure at upper levels can disrupt the entire system.
- Lower-level agents have limited independence.



Key takeaways

I hope you can:

- Grab the idea what AI agent is
- The main features of Azure AI Agent services
- How to implement a basic AI Agent using Azure AI Agent services
- The agentic architecture of AI Agents
- Multi agent systems



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Thank you

Nash Tech.